SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Sceanario Based Questions

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO4 -Sceanario Based Questions**	**Assessment:**	**Assessment Tool 1– Sceanario Based Questions**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO4 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
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Section / Batch:		**Date:**	13/08/2026

Code of Conduct

I ___POOJA S(192524211)___ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _______POOJA S__________________

1. A software company claims that its new application has an average response time of less than 2 seconds. A hypothesis test gives a p-value of 0.004. At $\alpha = 0.01$, determine whether there is sufficient evidence to support the company's claim.

Answer:

Step 1: State the Hypotheses

Let μ be the true average response time of the new application.

- **Null Hypothesis (H_0):** $\mu \geq 2$ seconds
- **Alternative Hypothesis (H_1):** $\mu < 2$ seconds

The company's claim is represented by the alternative hypothesis.

Step 2: Identify the Significance Level

Given:

- **Significance level (α) = 0.01**
- **p-value = 0.004**

Step 3: Compare the p-value with α

p-value = 0.004

$\alpha = 0.01$

Since:

0.004 < 0.01

the p-value is less than the significance level.

Step 4: Decision

Since the p-value is less than α, **reject the null hypothesis (H_0).**

Step 5: Conclusion

There is sufficient statistical evidence at the **1% significance level** to support the company's claim that the new application's average response time is **less than 2 seconds**.

Final Answer:

Since p-value (0.004) < α (0.01), we reject the null hypothesis. Therefore, there is sufficient evidence to conclude that the application's average response time is significantly less than 2 seconds, supporting the company's claim.

2. A digital marketing team launches a new advertisement and wants to know whether it increases the click-through rate. Before the campaign, the rate was 6%. After the campaign, a hypothesis test produces a p-value of 0.12. What should the data science team conclude at the 5% significance level?

Answer:

The purpose is to test whether the new advertisement has increased the click-through rate from the previous value of **6%**.

<u>Step 1: Formulate the Hypotheses</u>

- **Null Hypothesis (H_0):** The click-through rate has not increased above 6%.
  H_0: $p \leq 0.06$
- **Alternative Hypothesis (H_1):** The click-through rate is greater than 6%.
  H_1: $p > 0.06$

Therefore, this is a **right-tailed hypothesis test**.

<u>Step 2: Significance Level</u>

The given significance level is:

$\alpha = 0.05$

<u>Step 3: Compare p-value and α</u>

The hypothesis test gives:

p-value = 0.12

Since:

$0.12 > 0.05$

the p-value is larger than the chosen significance level.

<u>Step 4: Make the Decision</u>

Because **p-value > α**, we **fail to reject the null hypothesis (H_0)**.

<u>Step 5: Conclusion</u>

At the **5% significance level**, the available data does not provide sufficient statistical evidence to show that the new advertisement has increased the click-through rate.

<u>Final Answer:</u>

<u>Since the p-value of 0.12 is greater than the significance level of 0.05, we fail to reject H_0. Hence, there is insufficient evidence to conclude that the new advertisement significantly increases the click-through rate above 6%</u>.

3.A machine-learning team develops a new classification model. The old model has an accuracy of 82%, while the new model achieves 85%. A statistical test gives a p-value of 0.01. Explain what the p-value means and whether the team should conclude that the new model performs significantly better.

Answer:

Step 1: State the Hypotheses

The objective is to determine whether the new classification model performs better than the old model.

- **Null Hypothesis (H_0):** The new model does not perform better than the old model.
  H_0: Accuracy ≤ 82%
- **Alternative Hypothesis (H_1):** The new model performs better than the old model.
  H_1: Accuracy > 82%

This is a **right-tailed hypothesis test**.

Step 2: Understand the p-value

The given p-value is:

p-value = 0.01

A p-value of 0.01 means that, assuming the null hypothesis is true, there is a **1% probability of obtaining results as extreme as or more extreme than the observed result** due to random chance.

Step 3: Compare the p-value with the Significance Level

Using the commonly used significance level:

$\alpha = 0.05$

Since:

$0.01 < 0.05$

the p-value is less than the significance level.

Step 4: Decision

Since **p-value < α**, we **reject the null hypothesis (H_0)**.

Step 5: Conclusion

The new model has an accuracy of **85%**, compared with **82%** for the old model. Since the p-value is only **0.01**, the observed improvement is statistically significant at the 5% significance level.

Final Answer:

A p-value of 0.01 means that there is only a 1% chance of obtaining an improvement as extreme as the observed result if the new model were not actually better. Since 0.01 < 0.05, we reject the null hypothesis. Therefore, there is sufficient statistical evidence to conclude that the new classification model performs significantly better than the old model.

4. A factory claims that its defect rate is 2%. After randomly inspecting products, a 95% confidence interval for the defect rate is found to be (2.4%, 3.6%). At the same time, a hypothesis test gives a p-value of 0.01. Interpret both results and explain whether the company's claim should be rejected.

Answer:

Step 1: State the Company's Claim

The factory claims that its defect rate is **2%**.

Therefore:

- **Null Hypothesis (H_0):** p = 0.02
- **Alternative Hypothesis (H_1):** p ≠ 0.02

This is a **two-tailed hypothesis test**.

Step 2: Interpret the 95% Confidence Interval

The given 95% confidence interval is:

(2.4%, 3.6%)

This means that we are **95% confident that the true defect rate lies between 2.4% and 3.6%**.

The company's claimed defect rate of **2%** is **not included** in this confidence interval.

Therefore, the sample data suggests that the actual defect rate is likely **higher than 2%**.

Step 3: Interpret the p-value

The given:

p-value = 0.01

Using a significance level of:

$\alpha = 0.05$

Compare:

$0.01 < 0.05$

Since the p-value is less than the significance level, we **reject the null hypothesis (H_0)**.

Step 4: Decision

Both the confidence interval and p-value provide the same conclusion:

- **2% is outside the 95% confidence interval.**
- **p-value (0.01) < α (0.05).**

Therefore, there is sufficient statistical evidence against the company's claim.

Step 5: Conclusion

The company's claim that the defect rate is **2% should be rejected**. The 95% confidence interval suggests that the actual defect rate lies between **2.4% and 3.6%**, which is higher than the claimed rate.

Final Answer:

Since the claimed defect rate of 2% does not fall within the 95% confidence interval of (2.4%, 3.6%), and the p-value of 0.01 is less than the significance level of 0.05, we reject the null hypothesis. Therefore, there is sufficient evidence to conclude that the actual defect rate is significantly different from, and likely higher than, 2%.

5.An online shopping company wants to determine whether a new user-interface design increases the average amount spent per customer. The data scientist calculates a 95% confidence interval for the difference in average spending as (₹120, ₹480) and obtains a p-value of 0.002. Interpret the confidence interval and p-value, and provide a business recommendation.

Answer:

Step 1: State the Hypotheses

The company wants to determine whether the new user-interface design increases the average amount spent per customer.

- **Null Hypothesis (H_0):** The new UI does not increase average spending.
  H_0: $\mu_{new} - \mu_{old} \leq 0$
- **Alternative Hypothesis (H_1):** The new UI increases average spending.
  H_1: $\mu_{new} - \mu_{old} > 0$

This is a **right-tailed hypothesis test**.

Step 2: Interpret the 95% Confidence Interval

The 95% confidence interval for the difference in average spending is:

(₹120, ₹480)

This means that we are **95% confident that the true increase in average spending per customer lies between ₹120 and ₹480**.

Since the entire confidence interval is **greater than ₹0**, it indicates that the new UI design has a positive effect on customer spending.

Step 3: Interpret the p-value

The given:

p-value = 0.002

Using the significance level:

$\alpha = 0.05$

Compare:

$0.002 < 0.05$

Since the p-value is less than the significance level, we **reject the null hypothesis (H_0)**.

Step 4: Decision

Both the confidence interval and p-value support the same conclusion:

- The confidence interval **(₹120, ₹480)** does not contain ₹0.
- The entire interval is positive, indicating an increase in spending.
- The p-value of **0.002** is much smaller than **0.05**.

Therefore, the increase in average customer spending is **statistically significant**.

Step 5: Business Recommendation

The company should consider **implementing the new user-interface design**, as the results provide strong statistical evidence that it increases the average amount spent by customers.

However, before implementing it for all users, the company may conduct further testing to evaluate factors such as **implementation cost, customer satisfaction, and long-term business impact**.

Final Answer

The 95% confidence interval of (₹120, ₹480) indicates that the new UI is expected to increase average customer spending by between ₹120 and ₹480. Since the interval does not include ₹0 and the p-value of 0.002 is less than $\alpha = 0.05$, we reject the null hypothesis. Therefore, there is strong statistical evidence that the new UI design significantly increases average spending per customer, and the company should consider adopting the new design.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	
Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	